Multivariate Orthogonal Polynomials (Symbolically)

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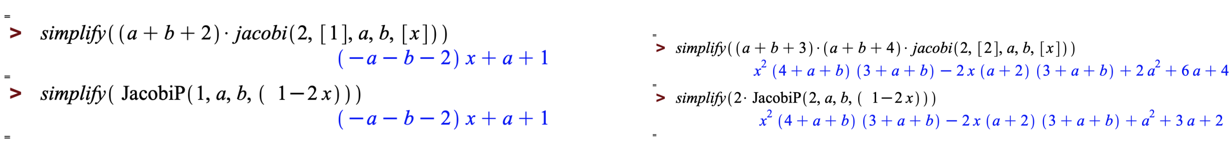
1 Introduction

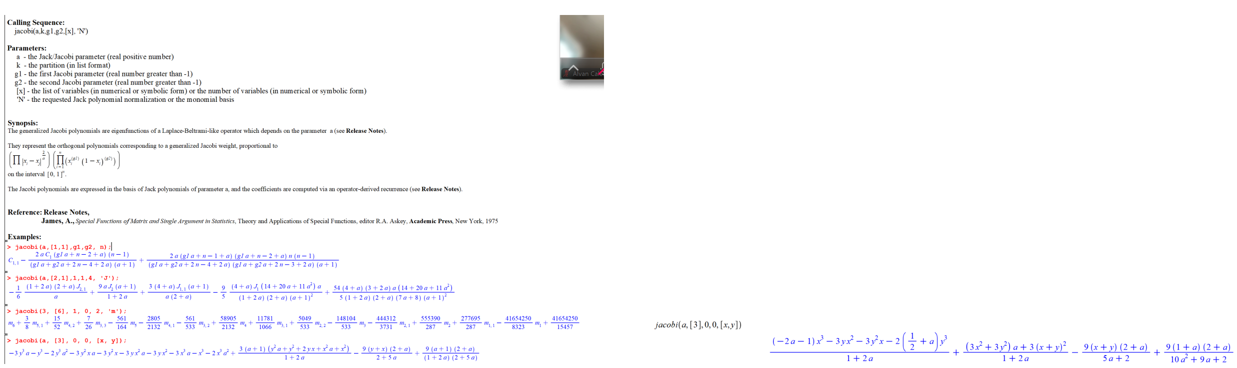
The motivation of this work is to resolve an error in the Maple implementation of [DES07]. The simplest proof of fault is for Jacobi polynomials of second order. Figure 1b shows that the MOPS implementation differs in the constant term from the built-in Maple implementation by a factor of 2.



(a) Linear (Valid) (b) Quadratic (Invalid)

Figure 1: Jacobi Polynomial Error in MOPS (Maple)

Due to Maple licensing issues, I, Alvan, ported the Maple implementation to Julia. The hope was to enable easier debugging, understanding of the code, and unit testing for various functions to prevent regressions while tinkering with the implementation to find the error. Using the MOPS Maple '/Help Files', we were able to turn example usage into test cases. Luckily, since Duy Thuc did have access to Maple, we were able to verify these examples in Maple, finding a documentation bug.



(a) Help Page (b) Verification

Figure 2: Discrepancy in Fourth Example of Help Page for jacobi($\cdot$)

Through the process of unit testing the Julia implementation, I also found a typo in the paper. Table 1 of

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[DES07] lists the coefficient of m_3 in $J^{\alpha}_{[3]}$ as $(1+\alpha)(2+\alpha)$ when it should be $(1+\alpha)(1+2\alpha)$, as demonstrated by the following.

```
julia> using MOPS; using SymPy; @syms a;

julia> Jack(a, [3], :J)
(a + 1)⋅(2⋅a + 1)⋅m[3] + (3⋅a + 3)⋅m[2, 1] + 6⋅m[1, 1, 1]
```

Figure 3: Coefficient of m_3 in $J^{\alpha}_{[3]}$

With all tests passing, the Julia port was able to reproduce the Jacobi constant term error.

```
julia> simplify((a+b+3)*(a+b+4)*Jacobi(2//1, [2], a, b, [x]))
 2
x ⋅(a + b + 3)⋅(a + b + 4) - 2⋅x⋅(a + 2)⋅(a + b + 3) + 2⋅(a + 1)⋅(a + 2)

julia> simplify((a+b+2)*Jacobi(2//1, [1], a, b, [x]))
a - x⋅(a + b + 2) + 1
```

Figure 4: Reproduction of Figure 1b in Julia

As hinted by Sungwoo, we started with looking at the implementation of the generalized binomial coefficients. While the recursion to the contiguous coefficients checked out, the computation of the contiguous coefficients did not. From (9) of [DES07],

$$\binom{\sigma^{(i)}}{\sigma} = j_{\sigma}^{-1} g^{\sigma^{(i)}}_{\sigma\,1}, \quad g^{\sigma^{(i)}}_{\sigma\,1} = \left(\prod_{s\in\sigma} A_{\sigma^{(i)}}\right)\left(\prod_{s\in\sigma} B_{\sigma^{(i)}}\right)$$

where $(A_{\sigma^{(i)}}, B_{\sigma^{(i)}}) = (h^*_{\sigma}(s), h^{\sigma^{(i)}}_*)$ if s is in the i-th column of σ and is $(h^{\sigma}_*(s), h^{\sigma^{(i)}}_*)$ otherwise. However, 'MOPS/GBC/cont' did not take this implementation. It instead computed a simpler product with no reference to upper and lower hook lengths at all. Restoring the implementation to one faithful to (9) of [DES07], titled `GBC_cont_explicit(*)` in the Julia port, I found that the original efficient implementation was indeed wrong.

```
julia> using MOPS; using SymPy; @syms a;
Precompiling MOPS
  1 dependency successfully precompiled in 2 seconds. 32 already precompiled.

julia> GBC_cont(a, [3], 1)
3

julia> GBC_cont_explicit(a, [3], 1)
3⋅a   3
─── + ─
 2    2
```

Figure 5: Discrepancy between Original and Explicit Implementations of $\binom{\sigma^{(i)}}{\sigma}$ for $\sigma^{(i)} = (3)$ and $\sigma = (2)$

One can verify that the explicit implementation is indeed correct by manual computation.

$$\binom{\sigma^{(i)}}{\sigma} = \frac{\left(\prod_{s\in\sigma} A_{\sigma^{(i)}}\right)\left(\prod_{s\in\sigma} B_{\sigma^{(i)}}\right)}{j_{\sigma}} = \frac{((1+\alpha)\cdot\alpha)\,(3\alpha\cdot(1+\alpha))}{(2\alpha\cdot\alpha)\,((1+\alpha)\cdot 1)} = \frac{3\alpha^2(1+\alpha)^2}{2\alpha^2(1+\alpha)} = \frac{3(1+\alpha)}{2}$$

The explicit implementation sufficed for the rest of our testing. However, a simpler implementation is possible by noticing that the only cells of the Young diagram which contribute to the product are those in the same column or row of the cell added, between $\sigma^{(i)}$ and σ. However, even with this fix, the output of Figure 4 remained exactly the same. This is because the generalized binomial coefficients involved in the computation of `Jacobi(2, [2], a, b, [x])` are $\binom{(2)}{(2)}, \binom{(2)}{(1)}, \binom{(2)}{()}$ which are easily resolved by base cases.

From Section 2.7.2 of [DES07], the Jacobi polynomials are given by

$$J_\kappa^{\alpha,g_1,g_2}(x_1,\dots,x_m) = \left(g_1 + \frac{m-1}{\alpha} + 1\right)_\kappa C_\kappa^\alpha(I_m) \sum_{\sigma\subseteq\kappa} (-1)^{|\sigma|} \frac{c_{\kappa\sigma}^\alpha}{\left(g_1 + \frac{m-1}{\alpha} + 1\right)_\sigma} \cdot \frac{C_\sigma^\alpha(x_1,\dots,x_m)}{C_\sigma^\alpha(I_m)} \tag{1}$$

where the "Jacobi coefficients", $c_{\kappa\sigma}^\alpha$, are given by the recurrence relation

$$c_{\kappa\sigma}^\alpha = \frac{1}{\left(g_2 + g_1 + \frac{2}{\alpha}(m-1) + 2\right)(k-s) + \rho_\kappa^\alpha - \rho_\sigma^\alpha} \sum_{i \text{ allowable}} \binom{\kappa}{\sigma^{(i)}} \binom{\sigma^{(i)}}{\sigma} c_{\kappa\sigma^{(i)}}^\alpha \tag{2}$$

Via printline debugging, I found that the $\sigma = ()$ term of (1) in the Jacobi polynomial expansion is solely responsible for the constant term. Specifically, the additional factor of two in the constant term in Figure 4 comes from $c_{(2)()}^\alpha$; as we recur from $c_{(2)()}^\alpha \to c_{(2)(1)}^\alpha \to c_{(2)(2)}^\alpha$, we pick up two factors of $\binom{(2)}{(1)}$, first from $\binom{\kappa}{\sigma^{(i)}}$ and second from $\binom{\sigma^{(i)}}{\sigma}$, while only the denominator of the first recursion step is significant, canceling one factor.

By Duy Thuc's explicit calculation of the quadratic case, we were able to conclude that the Julia code was a faithful implementation of [DES07] but was still the wrong Jacobi polynomial. Unfortunately, this means that there is some deeper algorithmic error in [DES07] itself.

With some playing around, I found that if $\binom{\kappa}{\sigma^{(i)}}$ in the $c_{\kappa\sigma}^\alpha$ recurrence is omitted, we do not pick up these factors of 2 and we get the correct Jacobi polynomial for the quadratic case. However, if we move to $\kappa = [3]$, the polynomial no longer looks correct as it is clearly not orthogonal with the quadratic one.

2 Manual Computation of Quadratic Case

In the test case, for $\kappa = (2)$, the subpartitions are $(2), (1), (0)$. We have $k = |\kappa| = 2$ and $s = 0, 1, 2$. See that in the univariate case, Jack polynomials are just the standard monomials $C_{(r)}^\alpha(x) = x^r$ (since we divide by $C_\sigma^\alpha(I_m)$, the normalization choice for these should not matter). See that for $m = 1$, we have $\rho_{(r)}^\alpha = r(r-1)$. So $\rho_{(2)} = 2, \rho_{(1)} = 0, \rho_{(0)} = 0$. We have

$$\binom{(2)}{(2)} = 1, \quad \binom{(2)}{(1)} = 2, \quad \binom{(1)}{(0)} = 1, \quad \binom{(2)}{(0)} = 1 \tag{3}$$

We set $c_{(2),(2)} = 1$ and compute $c_{(2)(1)}$ an $c_{(2)(0)}$. First consider $c_{(2)(1)}$. Here $s = 1$ and $k - s = 1$. So the denominator in (2) becomes

$$(g_2 + g_1 + 2)(k-s) + \rho_{(2)} - \rho_{(1)} = (a + b + 2) + 2 - 0 = a + b + 4 \tag{4}$$

See that the only allowable i is $i = 1$, so that $\sigma^{(1)} = (2)$. Therefore, the coefficient is $\binom{2}{2} \cdot \binom{2}{1} c_{(2)(2)} = 2$. We have

$$c_{(2)(1)} = \frac{2}{a+b+4}$$

Now for $c_{(2)(0)}$, we have $s = 0, k - s = 2$ and so the denominator is given by

$$(a + b + 2) \cdot 2 + \rho_{(2)} - \rho_{(0)} = 2(a + b + 2) + 2 = 2(a + b + 3) \tag{5}$$

Again, the only allowable i is $i = 1$, which gives $\sigma^{(i)} = (1)$. Therefore, numerator is given by

$$\binom{2}{1} \cdot \binom{1}{0} c_{(2)(1)} = \frac{4}{a+b+4} \tag{6}$$

So we have

$$c_{(2)(0)} = \frac{1}{2(a+b+3)} \cdot \frac{4}{a+b+4} = \frac{2}{(a+b+3)(a+b+4)} \tag{7}$$

We have for the univariate case

$$J^{2,a,b}_{(2)}(x) = (a+1)_2 \sum_{s=0}^{2} (-1)^s \frac{c_{(2)(s)}}{(a+1)_s} x^s \tag{8}$$

So we have the following three terms from the sum:

$$\begin{aligned} s = 2: &\quad \frac{1}{(a+1)(a+2)} x^2 \\ s = 1: &\quad -\frac{2}{(a+b+4)(a+1)} x \\ s = 0: &\quad \frac{2}{(a+b+3)(a+b+4)} \end{aligned} \tag{9}$$

But we multiply the sum by $(a+1)_2 = (a+1)(a+2)$, so we have

$$J^{2,a,b}_{(2)}(x) = x^2 - \frac{2(a+2)}{a+b+4} x + \frac{2(a+1)(a+2)}{(a+b+3)(a+b+4)} \tag{10}$$

Multiplying by $(a+b+3)(a+b+4)$ gives

$$(a+b+3)(a+b+4)x^2 - 2(a+2)(a+b+3) + 2(a+1)(a+2) \tag{11}$$

which matches the MOPS output and not the built-in Maple output.

3 Code Availability

See https://github.com/arulandu/MOPS for both the original MOPS maple project in the `/maple` folder and the Julia port in the `/julia` folder. As a disclaimer, the port is still sloppy with AI generated code.

References

[DES07] Ioana Dumitriu, Alan Edelman, and Gene Shuman. Mops: Multivariate orthogonal polynomials (symbolically). *Journal of Symbolic Computation*, 42(6):587–620, 2007. Also available at arXiv:math-ph/0409066.